

PROFILE

Interdisciplinary PhD researcher working at the intersection of **data science, geospatial AI, environmental modeling, and public health**. Experienced in building ML/DL and statistical models from complex spatiotemporal datasets, including Earth observation, environmental sensors, meteorological records, health data, and socio-economic indicators. Develops end-to-end research workflows from data engineering and modeling to interactive tools used by community and government partners.

EXPERIENCE

PhD Researcher: Urban Climate

[Python, R, PostgreSQL, Google Earth Engine, QGIS/ArcGIS, Tableau]

Aug 2023 – Present

Healthy Cities Lab, Indiana University

Bloomington, IN

- **Geospatial AI & Earth Observation:** Develop ML/DL workflows for urban environmental modeling using **Landsat, Sentinel, NAIP, Google Earth Engine**, and high-resolution urban-form data. Built Local Climate Zone classification models using tree-based methods and CNNs, and currently explore **Earth Observation foundation-model embeddings** for transferable urban representations.
- **Spatiotemporal Temperature Modeling:** Develop machine-learning and deep-learning models to predict hyperlocal near-surface air temperature by integrating **satellite imagery, urban morphology, meteorological data, and in-situ sensors**; run large-scale Python modeling workflows on university HPC infrastructure.
- **Climate-Health & Statistical Modeling:** Integrated electronic health records with daily meteorological data to quantify health risks during extreme heat. Used **GLMs and estimated marginal means (emmeans)** to compare emergency-department visit risk during heatwave and control periods, and **Distributed Lag Non-Linear Models (DLNMs)** to characterize non-linear, lagged temperature-health relationships and identify temperatures associated with elevated ED risk.
- **Environmental Sensing & Decision Support:** Used geospatial classification and urban-form analysis to guide deployment of distributed environmental sensors and analyzed resulting microclimate data. Built **interactive maps and dashboards** using Flask, Chart.js, Plotly/Dash, and ArcGIS to communicate heat exposure and vulnerability to cities and community partners.
- **Geospatial Data Engineering:** Build reproducible pipelines to acquire, clean, integrate, and analyze multi-source **satellite, sensor, climate, health, and socio-economic data** using Python/R, PostgreSQL/PostGIS, Rasterio, GeoPandas, Xarray, and cloud/HPC computing.

Software Engineering Intern

[C#, .NET MVC, Fullstack Web Development, Jira]

May 2022 – Dec 2022

Bloom Insurance

Bloomington, IN

- Developed **30+ production features, APIs, and bug fixes** in C#/.NET and improved application performance while working in an Agile development team.

Software Developer

[Falcon API, HashiCorp Vault, Elasticsearch, Grafana, Terraform, OpenStack]

Nov 2020 – Aug 2021

National Payments Corporation of India

Hyderabad, India

- Built a fully containerized Python API and centralized key-management system using **HashiCorp Vault**, delivered within 2 months.
- Developed automated infrastructure and data workflows using **Docker, Terraform, GitLab CI/CD, Linux, and OpenStack**, providing experience deploying reproducible computational systems.

EDUCATION

PhD in Informatics

May 2027 (expected), GPA 3.9

Indiana University Bloomington

Bloomington, IN

Master's in Data Science

May 2023, GPA 3.9

Indiana University Bloomington

Bloomington, IN

SKILLS

- **Languages:** Python, R, SQL, C#, Bash, JavaScript
 - **ML / DL:** Scikit-learn, PyTorch, CNNs, Classification, Regression
 - **Earth Observation & GIS:** Google Earth Engine, QGIS, ArcGIS Pro, ArcPy, Landsat, Sentinel-2, NAIP, GDAL
 - **Statistics:** GLMs, DLNMs, emmeans, Time-Series Analysis, Spatial Interpolation, Exposure-Response Modeling
 - **Geospatial Python:** GeoPandas, Rasterio, Xarray, GDAL, Folium, Leaflet
- **Data Engineering:** PostgreSQL/PostGIS, ETL Pipelines, API Development (Flask), Multi-source Data Integration
 - **Visualization & Apps:** Plotly/Dash, Streamlit, Shiny, Chart.js, Tableau, Esri JavaScript SDK
 - **Cloud & Compute:** GCP, HPC/Supercomputing, Docker, Terraform, GitLab CI/CD
 - **Research Methods:** EO Foundation Models, Feature Engineering, Hyperparameter Tuning

PRESENTATIONS & PUBLICATIONS

- Habeeb, D., Subramanian, L., **Devajji, R.**, Polak, N. (2026) "Publics, Place, and Sensors: Co-Designing a High-Density Environmental Sensor Network with a Volunteer-Run Community Organization." *2026 CHI Conference*. doi:10.1145/3772318.3790740
- Habeeb, D., **Devajji, R.**, Wilson, J. (2026) "Identifying local climate zones in two U.S. Midwestern cities: a comparison of methods" *Computational Urban Science* doi:10.1007/s43762-026-00261-w
- Habeeb, D., **Devajji, R.**, et al. (2025). "Design and Evaluation of Wearable Solar Radiation Shields for Enhanced Personal Heat Exposure Monitoring." *Sensors*, 25(3), 945. doi:10.3390/s25030945
- Tu, H., **Devajji, R.**, Horan, T. (2025). "Algorithmic Literacy and Digital Privacy in the US: An Exploratory Study Using Data Visualization." *IEEE ICAD*. doi:10.1109/ICAD65464.2025.11114060
- **Devajji, R.** & Habeeb, D. (2025). "A Machine Learning Approach to Predicting Hyperlocal Temperature in the Built Environment." In *Association of Collegiate Schools of Planning (ACSP) Annual Conference*.
- **Devajji, R.**, Habeeb, D., & Wilson, J. S. (2025). "Extreme Heat and Health Impacts: A Study of Emergency Department Visits in Midwest Communities." In *American Association of Geographers (AAG) Annual Meeting*.
- Habeeb, D., Polak, N., **Devajji, R.** (2024). "Leveraging an Urban Environmental Sensing Network to Improve Extreme Heat Resilience." *IEEE IGSC*. doi:10.1109/IGSC64514.2024.00036